

AHMED HAITHAM

Cairo, Egypt

☎ +201156050380 ✉ ahmedhaithamegy@gmail.com  [linkedin.com/in/ahmed-haitham-mohamed](https://www.linkedin.com/in/ahmed-haitham-mohamed)
 github.com/AhmedHaitham911

About Me

Passionate Computer Science student with hands-on experience in full-stack web development, game design, data engineering, data science and system engineering. Seeking an internship or junior role to apply skills in a real-world environment and grow as a software developer.

Education

The British University in Egypt

Sep. 2022 – Jun 2026

Bachelor of Science in Computer Science

Cairo, Egypt

Relevant Coursework

- | | | | |
|--------------------------|-----------------------|------------------------|---------------------------|
| • Data Structures | • Cloud Computing | (Flutter) | • Database |
| • Analysis of Algorithms | • Theory Of Computing | • Computer Security | • Artificial Intelligence |
| • Data Mining And | • Compiler Design | • Modeling And | • Operating Systems |
| Warehousing | • Web Development | Simulation | • Computer Architecture |
| • Data Analysis And | (React) | • Software Engineering | • Embedded Systems |
| Data Visualization | • Mobile Development | • Computer Networks | • Distributed Systems |
| | | | • Computer Graphics |

Experience

BARQ Systems

August 2025– september 2025

Data Management Track Intern

- Designed and implemented ETL pipelines, handling missing values and automating processes with Microsoft fabric.
- Worked on real-world projects involving data preprocessing, analysis, and visualization.
- Strengthened skills in data cleaning, preprocessing, and advanced data handling.
- Conducted data analysis and visualizations using Python libraries such as Pandas, NumPy, and Matplotlib
- Strengthened problem-solving abilities and applied statistical methods to real datasets.
- Gained hands-on experience in Large Language Models (LLMs) and their practical applications.
- Learned to apply Retrieval-Augmented Generation (RAG) techniques for knowledge retrieval and AI-powered assistance.
- Strengthened understanding of data engineering workflows, including data preprocessing, transformation, and structuring for AI systems.
- Developed practical skills in interactive AI application development using Gradio.
- Improved skills in text embedding, vector databases (FAISS), and semantic search techniques.
- Enhanced proficiency with AI frameworks and tools such as LangChain, Hugging Face, and Transformers.
- Improved collaboration and professional communication while working within a real-world enterprise environment.

Projects

WE Enterprise RAG Assistant | *Python, RAG, ChromaDB, Hugging Face, Gradio*

August 2026

- * Built an AI pipeline to detect offensive and harmful Arabic text.
- * Analyzed content at both word and sentence level for explicit and contextual offenses.
- * Used multiple BERT models for improved accuracy and cross-validation.
- * Implemented a scoring system to classify content as safe, suspicious, or highly offensive.
- * Optimized for real-time processing using GPU acceleration.

Arabic Content Guard (Toxic Language) | *Hugging face, Bert*

August 2025

- * Built an AI pipeline to detect offensive and harmful Arabic text.
- * Analyzed content at both word and sentence level for explicit and contextual offenses.
- * Used multiple BERT models for improved accuracy and cross-validation.
- * Implemented a scoring system to classify content as safe, suspicious, or highly offensive.
- * Optimized for real-time processing using GPU acceleration.

Splunk chatbot | *Hugging face, python, gradio*

August 2025

- * Built a retrieval-based AI system for answering technical questions using LLMs and vector search (FAISS).
- * Processed and chunked PDF documentation into structured Markdown for NLP pipelines.

- * Generated semantic embeddings with HuggingFace Embeddings for efficient information retrieval.
- * Integrated conversation memory for context-aware responses.
- * Developed an interactive Gradio interface for real-time user queries.
- * Gained hands-on experience in Python, NLP, LLMs, Transformers, PDF parsing, and semantic search.

Amazon DeepSearch Assistant | *Python, Cohere API, LangChain, FAISS, Multi-Agent AI*

August 2025

- * Developed an AI-powered shopping assistant for conversational Amazon product search.
- * Designed a multi-agent workflow for query expansion, retrieval validation, and iterative refinement.
- * Implemented semantic search using FAISS and embeddings for accurate product retrieval.
- * Supported multilingual (English/Arabic) natural language queries.
- * Built a context-aware conversational interface using modern LLM and RAG techniques.

Healthcare Data Pipeline in Microsoft Fabric | *Microsoft Fabric, ELT, Galaxy schema*

August 2025

- * Developed an end-to-end healthcare data pipeline in Microsoft Fabric, transforming raw datasets into clean, analytics-ready tables.
- * Applied Medallion Architecture (Bronze → Silver → Gold) to structure and process both CSV and JSON healthcare data.
- * Designed a Galaxy Schema with Dimension and Fact tables, ensuring efficient and scalable analytics.
- * Automated data workflows using Fabric Pipelines, including ingestion, cleaning, transformation, and fact/dimension generation.
- * Optimized storage and query performance with Parquet and Delta Lake formats.
- * Documented the project and visualized the data model, demonstrating strong data engineering and reporting skills.

Travilla User Panel | *React, Vite, Tailwind CSS, Docker*

April 2025

- * Developed a user-friendly travel booking platform with React and Vite for fast and efficient front-end performance.
- * Implemented features like hotel and tour booking, user authentication, and payment processing.
- * Designed responsive UI components using Tailwind CSS and HeroUI for a seamless user experience.
- * Integrated protected routes for secure access to user-specific pages and functionalities.
- * Dockerized the application for easy deployment and scalability, with a docker-compose setup for local development.
- * Enhanced user engagement with animations, carousels, and dynamic content using Framer Motion and React Slick.

Fresh Cart | *React, Vite, TailwindCSS, Axios, React Router*

March 2025

- * Built a modern single-page e-commerce application using React and Vite for fast development and performance.
- * Developed features like user authentication, product browsing, shopping cart functionality, and order summary.
- * Implemented form validation with Formik and Yup, and integrated Toast notifications for user feedback.
- * Used TailwindCSS for responsive and visually appealing UI, and Axios for API communication with a backend.
- * Integrated protected routes and context-based authentication to enhance security and user experience.

Cinema Management System | *SQL*

November 2023

- * Developed a system to manage cinema operations including employees, branches, halls, films, ticket booking, and payments.
- * Handled employee roles, salaries, departments, and branch assignments.
- * Managed films by genre, language, schedule, and hall type (e.g., VIP, accessible).
- * Implemented ticket reservation with unique seat numbers, pricing logic, and customer info storage.
- * Integrated a payment system with invoice generation and support for multiple payment methods.

Flight Management System | *Java*

November 2023

- * Developed a global flight booking system with multi-role access and international support.
- * Implemented multi-role access with customized functionalities for passengers and employees.
- * Passengers can book/cancel flights, manage inquiries, and handle their account.
- * Employees can manage flights and user accounts through add/delete operations.
- * Streamlined booking operations and improved user experience through a clear and role-specific interface.

Smart Menu System | *SQL, Java*

November 2023

- * Developed an automated restaurant management system replacing traditional human roles with intelligent software.
- * Designed a Smart Menu for customers to browse dishes, place/edit/delete orders, request assistance, and submit feedback.
- * Implemented role-based functionalities for staff:
 - Chef: Manage and update the menu (add, modify, replace dishes).
 - Receptionist: Handle live inquiries and communicate with the manager.
 - Manager: Manage offers and user access control.
- * Enhanced customer experience and optimized restaurant operations by automating key services.

Technical Skills

Languages/Technologies/Frameworks/Platforms: Java, OOP, C++, C, C#, SQL, Python, Sass, Tailwind CSS, HTML/CSS, JavaScript, PostCSS, TypeScript, Git, GitHub, React, Unity, CISCO Packet Tracer, Ubuntu, Blender, Microsoft Fabric, ETL & ELT, Schema Design, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Apache Airflow, Power BI, Plotly, PostgreSQL.

Soft Skills : Communication, Teamwork, Adaptability, Time Management, Attention to Detail, Creativity, Critical Thinking, Problem-solving